Methodology Passport Card

1. Project information

Title	Robustness of Text-Based Phishing Detectors to LLM Paraphrasing
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Repository	https://github.com/Dias-M/ResearchMetod-MoldassanovDias-BilalSamat

2. Research type

Quantitative. The study is a controlled computational experiment with a paired design: the same phishing e-mails are scored by the same detector in their original and paraphrased form, and the outcome is a numeric metric (recall at a fixed false-positive rate). The question is "how much", so it is answered with effect sizes, confidence intervals and significance tests rather than with interviews or text analysis.

3. Hypotheses

Main hypothesis (RQ1, tested per detector: TF-IDF + LR, DistilBERT, zero-shot LLM):

- **H0:** LLM paraphrasing does not change recall at FPR = 1%: recall(original) − recall(paraphrased) = 0.
- **H1:** Paraphrasing lowers recall: recall(original) − recall(paraphrased) > 0.

Causal hypothesis (RQ2, controlling for detector, dataset and e-mail length):

- **H0:** β(paraphrased) = 0 in a logistic regression of the detection outcome.
- **H1:** β(paraphrased) < 0.

Optimisation hypothesis (RQ3):

- **H0:** After training on paraphrased e-mails, recall stays more than 2 percentage points (p.p.) below the baseline, or FPR rises by more than 1 p.p.
- **H1:** Recall is within 2 p.p. of the baseline and FPR rises by at most 1 p.p.

4. Variable matrix

Type	Variable	Values / unit
Independent	Text version	original / LLM-paraphrased
Independent	Detector	TF-IDF + logistic regression, DistilBERT, zero-shot LLM
Independent	Training regime (RQ3)	plain / augmented with paraphrases
Dependent	Recall at FPR = 1% (primary)	share of phishing e-mails detected, 0–1

Type	Variable	Values / unit
Dependent	F1, precision, ROC-AUC, realised FPR	0–1
Controlled	Datasets and train/validation/test split	fixed split, one seed for the split
Controlled	Decision threshold rule	set on validation ham of the original data, reused for all conditions
Controlled	Paraphraser	one open-weights LLM, fixed prompt, temperature and seed (`config/config.yaml`)
Controlled	Random seeds	42–46 for model initialisation and batch order
Controlled	Software	Python 3.11.9, Docker image `python:3.11.9-slim`, exact library versions in `requirements.txt`
Controlled	Hardware	CPU/GPU model recorded in `results/benchmark.log` for every run
Covariate	E-mail length, dataset source	characters; categorical

5. Metrics and baseline

Primary metric. Recall at FPR = 1%. A mail filter must keep false alarms rare, so detectors are compared at a fixed, low false-positive rate. The threshold is chosen on validation ham of the original data and reused for the paraphrased data, so only the phishing text differs between conditions.

Secondary metrics. F1, precision, ROC-AUC.

Guardrail metrics (a run is valid only if all hold; thresholds are proposed and may be adjusted with the supervisor):

Guardrail	Limit
Realised FPR on test ham	≤ 2% (target 1%)
Paraphrase acceptance by the quality filter	≥ 70% of phishing e-mails; 100% of URLs preserved
Seed-to-seed standard deviation of recall (DistilBERT)	≤ 2 p.p.
Pipeline integrity	`python src/benchmark.py` finishes with `BENCHMARK OK`, no errors

Baseline. Each detector's recall on the original (unparaphrased) test e-mails is the reference for its own drop. TF-IDF + logistic regression trained on original data is the classical baseline model (B0 in the README). RQ3 compares the augmented detectors with the plain ones on the same paraphrased test set.